H2 Math Summary

Notes

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1 Systems of Linear Equations

1.1 Forming the Equations

We read the context of the question and form equations depending on the context of the question.

Example 1.1

$$\left.\begin{aligned} a+b+c&=176 && —(1)\\ 0.85a+0.90b+0.95c&=161.80 && —(2)\\ 0.93a+0.95b+0.98c&=169.56 && —(3)\end{aligned}\right\} \Rightarrow a=32,\ b=44,\ c=100$$

Example 1.2

The "variables" may be expressions. To find x, y and z, use the given information to form the equations:

$$\left.\begin{aligned} 2x^3+3\ln y-\mathrm{e}^z&=18 && —(1)\\ x^3-\ln y+2\mathrm{e}^z&=9 && —(2)\\ x^3+2\ln y+\mathrm{e}^z&=11 && —(3)\end{aligned}\right\} \Rightarrow x^3=8,\ \ln y=1,\ \mathrm{e}^z=1$$

It is technically still 'linear' because we can think of $A=x^3$, $B=\ln y$, $C=\mathrm{e}^z$, which turns it into a linear system in A, B and C. A truly nonlinear equation could look like $x^2+x+y^2+2y-3z^3-xz+xy=1$; it cannot be made linear this way, because x appears as both x^2 and x, and there are product terms such as xz and xy.

1.2 Special Case: Number of Equations > Number of Variables

Sometimes the number of equations is more than the number of variables, and the solution must satisfy all of the equations.

Example 1.3

$$\left.\begin{aligned} \lambda+\mu&=1 && —(1)\\ 2\lambda-\mu&=5 && —(2)\end{aligned}\right\} \Rightarrow \lambda=2,\ \mu=-1$$

$$\lambda+3\mu=-1 \qquad —(3)$$

Sometimes there may be no solution, e.g. skew lines in vectors. In this case, solve for λ and μ using equations (1) and (2). Then sub into (3).

> **Note**
>
> Check (3): if LHS = RHS, the equations are consistent; if LHS ≠ RHS, there is no solution.

1.3 Special Case: Number of Equations < Number of Variables

Sometimes the number of equations is less than the number of variables, so there are infinitely many solutions.

Example 1.4

$$\left.\begin{aligned} x+y+z=6 \quad &—(1) \\ x-y+2z=5 \quad &—(2) \end{aligned}\right\} \Rightarrow x=\tfrac{11}{2}-\tfrac{3}{2}z,\ y=\tfrac{1}{2}+\tfrac{1}{2}z,\ z=z$$

> **Note**
>
> Useful trick: in contextual questions, the variables are usually non-negative integers, $z \in \mathbb{Z}_0^+$, so use this to find the solution from the general solution.

When $z=z$, it means that there are infinitely many solutions, and z can be any real value. So just sub that into x and y to get a correct solution.

$z=-1: (7,0,-1)$ $z=0: (\tfrac{11}{2},\tfrac{1}{2},0)$ $z=1: (4,1,1)$ $z=2: (\tfrac{5}{2},\tfrac{3}{2},2)$

$z=3: (1,2,3)$ $z=4: (-\tfrac{1}{2},\tfrac{5}{2},4)$ $z=5: (-2,3,5)$ $z=6: (-\tfrac{7}{2},\tfrac{7}{2},6)$

We can use the GC table to find all these solutions.

2 Inequalities

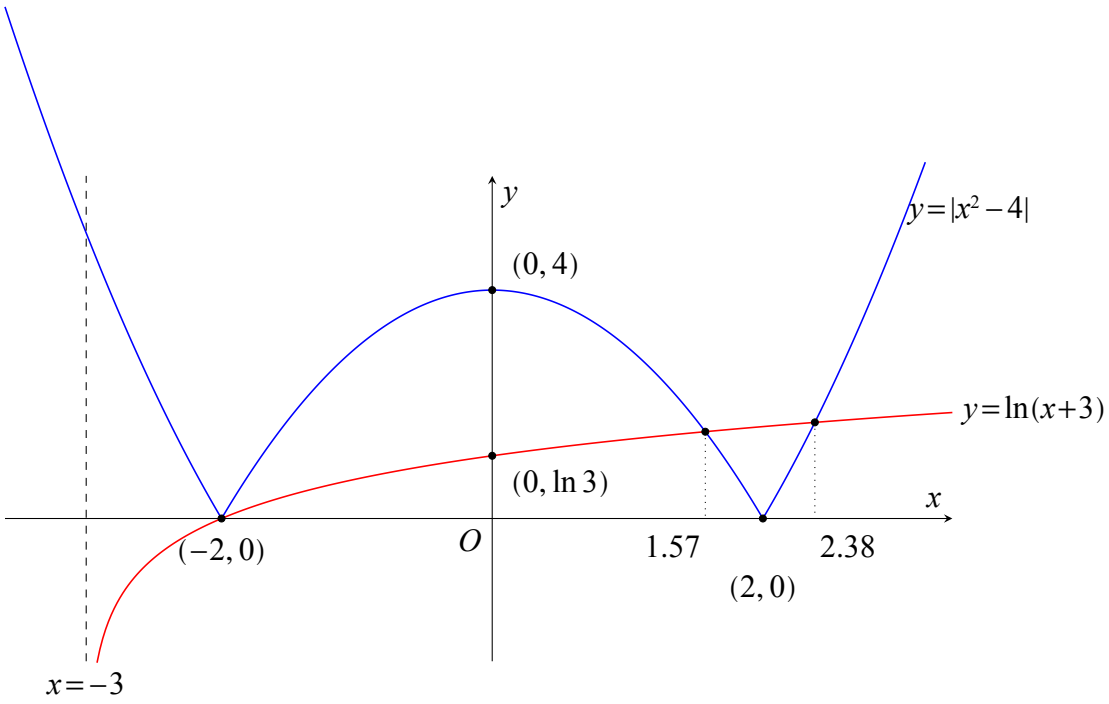
2.1 Solving Inequalities Graphically

Some inequalities cannot be solved algebraically, so we sketch the graphs and read off the solution.

- Rearrange to $f(x) > 0$, or compare $y = f(x)$ with $y = g(x)$.
- Sketch on the GC; mark the intersections and the asymptotes.
- Read off the x-values where the required graph is above or below.

Example 2.1

$$|x^2 - 4| \leqslant \ln(x+3)$$



Using GC, intersections at $x = 1.5746$, $x = 2.3840$

$$\therefore x = -2 \text{ or } 1.57 \leqslant x \leqslant 2.38$$

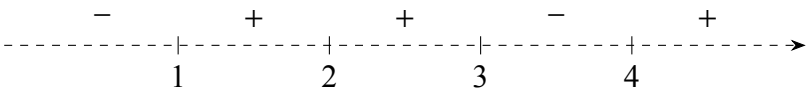
2.2 Sign Test Technique

We solve rational inequalities by finding where the expression changes sign.

- Make the right-hand side 0.
- Do **not** cross multiply unless the expression or bracket is always positive.
- Expressions that are always positive can be divided across, e.g. a quadratic with discriminant <0 and positive leading coefficient: $x^2+2x+5=(x+1)^2+4>0$.

Example 2.2

$$\frac{(x-1)(x-2)^2}{(x-3)(x-4)} \leqslant 0$$



$x=2$: numerator $=0$ (included)

$x=3,\ x=4$: denominator $=0$ (excluded)

$\therefore x \leqslant 1$ or $x=2$ or $3<x<4$

You can sketch the graph to check the answer.

> **Note**
>
> Crossing a factor with an odd power changes the sign; crossing a factor with an even power keeps the sign the same.

2.3 Common Substitutions

Once an inequality in x is solved, we can replace x with an expression to deduce the solution of a related inequality.

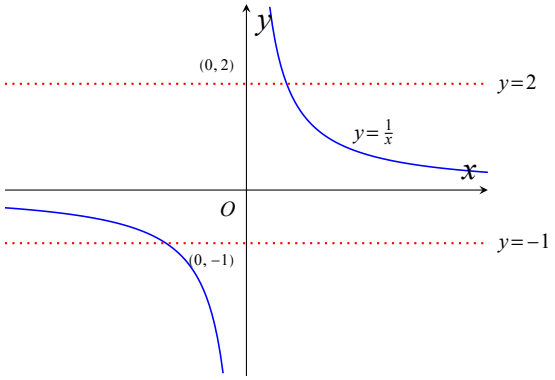
Replace x by	**Original inequality**		
	$x>2$	$x<-1$	$-1<x<2$
e^x	$x>\ln 2$	no solution	$x<\ln 2$
$\ln x$	$x>e^2$	$0<x<e^{-1}$	$e^{-1}<x<e^2$
$-x$	$x<-2$	$x>1$	$-2<x<1$
$x+1$	$x>1$	$x<-2$	$-2<x<1$
$\lvert x\rvert$	$x<-2$ or $x>2$	no solution	$-2<x<2$
x^2	$x<-\sqrt{2}$ or $x>\sqrt{2}$	no solution	$-\sqrt{2}<x<\sqrt{2}$
$\frac{1}{x}$	$\left(0<x<\frac{1}{2}\right)$	$(-1<x<0)$	$\left(x<-1 \text{ or } x>\frac{1}{2}\right)$
$\cos x,\ 0\leqslant x\leqslant 2\pi$	(no solution)	(no solution)	$(0\leqslant x<\pi \text{ or } \pi<x\leqslant 2\pi)$

Regardless of whatever substitution you use, using a graph will always work, but some can be solved without a graph.

For $\frac{1}{x}$, $\sin x$ and $\cos x$, draw the graph to solve.

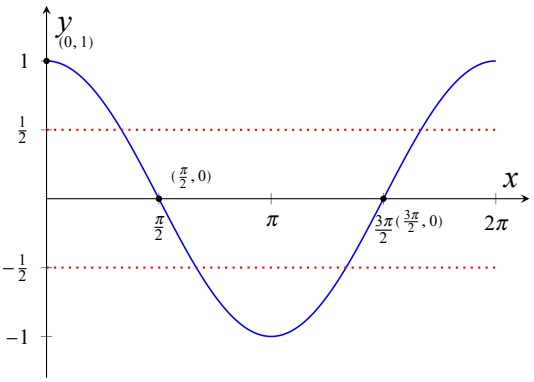
Example 2.3

$$\frac{1}{x}>2, \quad \frac{1}{x}<-1, \quad -1<\frac{1}{x}<2$$



Example 2.4

$$\cos x>\tfrac{1}{2}, \quad \cos x<-\tfrac{1}{2},$$
$$-\tfrac{1}{2}<\cos x<\tfrac{1}{2}, \quad 0\leqslant x\leqslant 2\pi$$

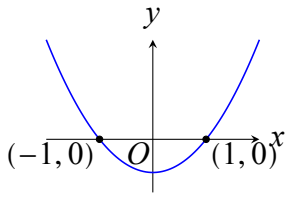


3 Graphing Techniques

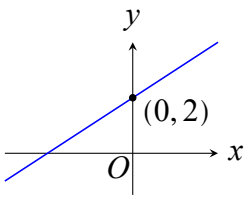
3.1 Key Features of a Curve

Key features of a curve:

(a) ***x*-intercepts:** let $y=0$.
e.g. $y=x^2-1$: $x^2-1=0 \Rightarrow x=\pm 1$.

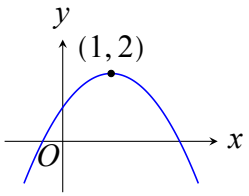


***y*-intercepts:** let $x=0$.
e.g. $y=x+2$: $y=2$.

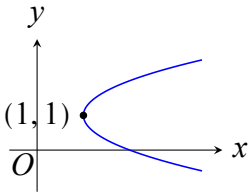


(b) **Stationary points**, when $\frac{dy}{dx}=0$ or $\frac{dy}{dx}$ is undefined. Label upon request.

Horizontal turning point: $\frac{dy}{dx}=0$ (tangent horizontal).
e.g. $y=-(x-1)^2+2$, maximum point $(1,2)$.

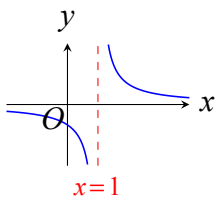
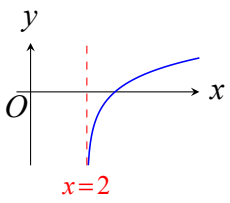
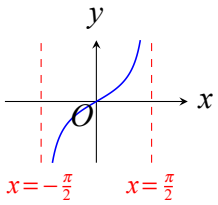


Vertical turning point: $\frac{dy}{dx}$ undefined (tangent vertical).
e.g. $x=(y-1)^2+1$, turning point $(1,1)$.



(c) **Asymptotes**

(i) **Vertical asymptotes.** If as $x \to k, y \to \pm\infty$, then $x = k$ is a vertical asymptote.

Common Occurrence	**Condition**	**Example and asymptote**	**Graph**
Fractions	denominator $= 0$	$y = \frac{1}{x-1}$ $x - 1 = 0 \Rightarrow x = 1$ $\therefore x = 1$ is a vertical asymptote	
Logarithmic graphs	inside ln $= 0$	$y = \ln(x-2)$ $x - 2 = 0 \Rightarrow x = 2$ $\therefore x = 2$ is a vertical asymptote	
Tangent graphs	inside tan $= \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$	$y = \tan x$ $x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$	

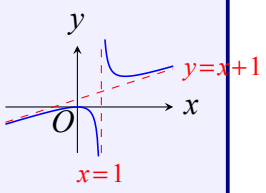
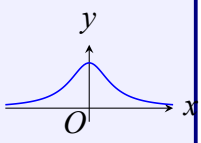
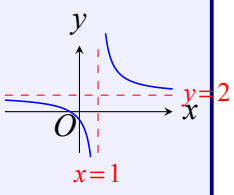
(ii) **Horizontal asymptotes.** As $x \to \pm\infty$, if $y \to k$, then $y = k$ is a horizontal asymptote.

- To find horizontal asymptotes, one should sub in $x \to \pm\infty$ into the equation. This is the only foolproof way to do it.
- A curve **CAN** intersect its horizontal asymptote.

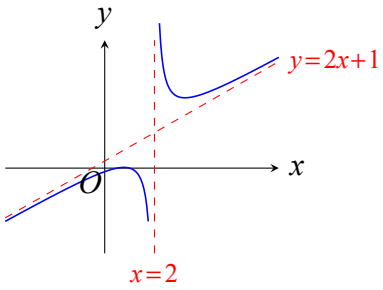
Note: there are more cases of horizontal asymptotes, but these are the common cases.

Useful trick to find horizontal asymptotes

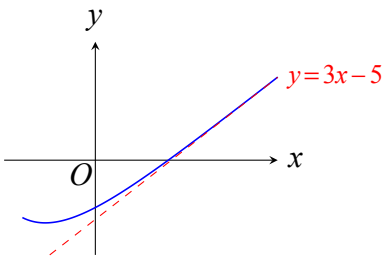
For rational functions $\frac{P(x)}{Q(x)}$:

Case	Horizontal asymptote	Example	Graph
numerator power > denominator power	no horizontal asymptote	$y=\frac{x^2}{x-1}$	
numerator power < denominator power	$y=0$	$y=\frac{1}{x^2+1}$	
numerator power = denominator power	ratio of coefficients of highest power	$y=\frac{2x+1}{x-1}$ $y=2$	

(iii) **Oblique asymptotes.** An asymptote of the form $y=mx+c$, $m\neq 0$.



e.g. $y=2x+1+\frac{3}{x-2}$



e.g. $y=3x-5+e^{-x}$

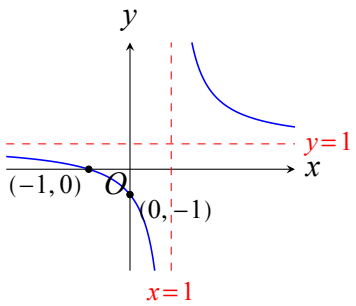
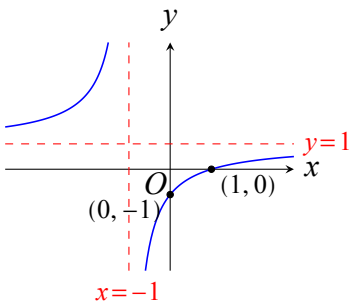
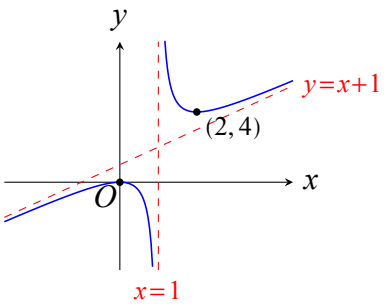
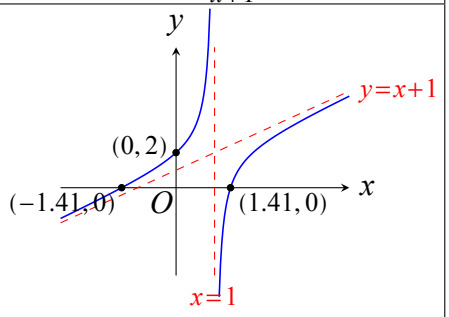
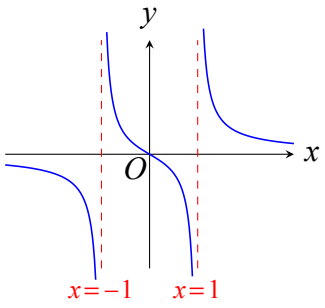
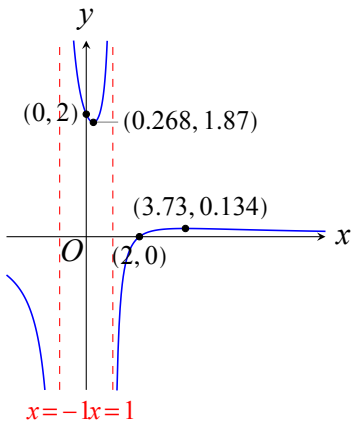
Note

For parametric curves, oblique asymptotes (if they exist) can only be found via the cartesian equation.

Graphs with oblique asymptotes come in the form $y=mx+c+g(x)$, where $g(x)\to 0$ as $x\to\pm\infty$; the oblique asymptote is $y=mx+c$.

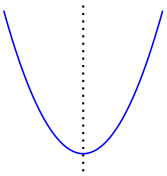
3.2 Graphs of Rational Functions

Two common shapes for each type.

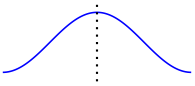
Type and form	Shape 1	Shape 2
Linear / linear $y=\frac{Ax+B}{Cx+D}$ Horizontal asymptote: $y=\frac{A}{C}$ Vertical asymptote: $x=-\frac{D}{C}$	 $y=\frac{x+1}{x-1}$	 $y=\frac{x-1}{x+1}$
Quadratic / linear $y=\frac{ax^2+bx+c}{dx+e}$ $=Ax+B+\frac{C}{dx+e}$ Oblique asymptote: $y=Ax+B$ Vertical asymptote: $x=-\frac{e}{d}$	 2 turning points $y=x+1+\frac{1}{x-1}\quad\left(\frac{Cd}{A}>0\right)$	 No turning points $y=x+1-\frac{1}{x-1}\quad\left(\frac{Cd}{A}<0\right)$
Linear / quadratic $y=\frac{ax+b}{cx^2+dx+e}$ Horizontal asymptote: $y=0$ Vertical asymptotes: solve $cx^2+dx+e=0$ (possibly no roots)	 $y=\frac{x}{(x-1)(x+1)}$	 $y=\frac{x-2}{(x-1)(x+1)}$

3.3 Symmetry

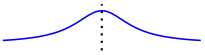
Even functions: $\mathrm{f}(-x)=\mathrm{f}(x)$. The graph is symmetric about the y-axis (line of symmetry $x=0$, dotted).



e.g. $y=x^2$



e.g. $y=\cos x$

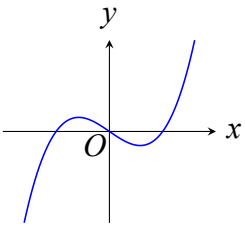


e.g. $y=\dfrac{1}{x^2+1}$

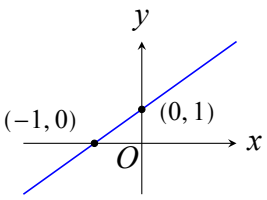
Odd functions: $\mathrm{f}(-x)=-\mathrm{f}(x)$.

The graph is symmetric about the origin.

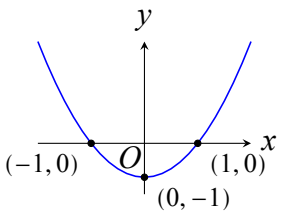
e.g. $y=x^3-x$



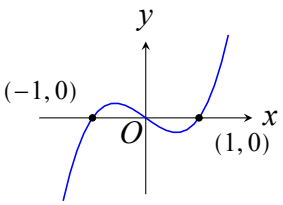
3.4 Basic Graphs



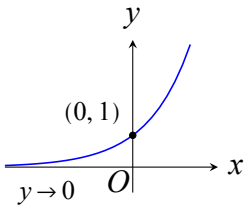
Linear $y=x+1$



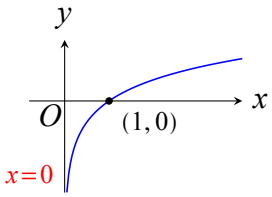
Quadratic $y=x^2-1$



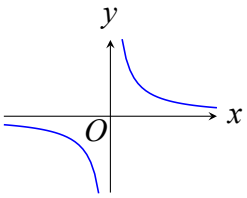
Cubic $y=x^3-x$



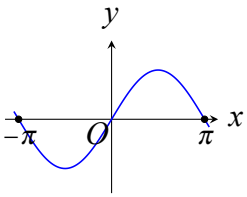
Exponential $y=e^x$



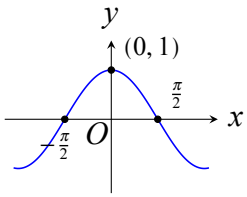
Logarithm $y=\ln x$



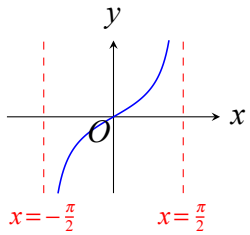
Reciprocal $y=\frac{1}{x}$



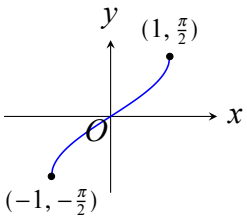
$y=\sin x$



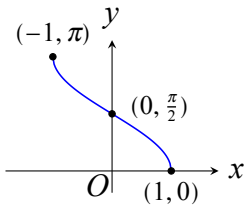
$y=\cos x$



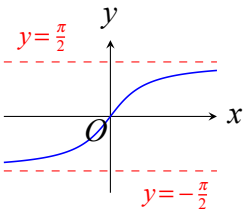
$y=\tan x$



$y=\sin^{-1}x$



$y=\cos^{-1}x$



$y=\tan^{-1}x$

3.5 Parametric Equations

A curve C can be described by giving both coordinates in terms of a parameter t:

$$x=x(t), \quad y=y(t), \quad t\in T$$

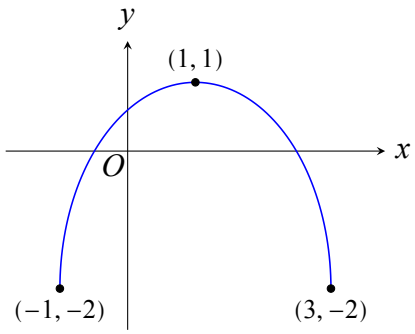
Example.

$x=2\cos t+1, \quad y=3\sin t-2, \quad 0\leqslant t\leqslant \pi$

$t=0$: $(3,-2)$ $\quad t=\frac{\pi}{2}$: $(1,1)$ $\quad t=\pi$: $(-1,-2)$

$x=0$: $\left(0, \frac{3\sqrt{3}}{2}-2\right)$

$y=0$: $\left(1\pm\frac{2\sqrt{5}}{3}, 0\right)$



CAUTION: sketching from parametric equations

ALWAYS take note of the range of t when sketching graphs. The restriction on t decides which part of the curve exists: above, $0\leqslant t\leqslant \pi$ gives only the upper half of the ellipse, with endpoints $(3,-2)$ and $(-1,-2)$, not the whole ellipse.

3.5.1 Finding the Cartesian Equation

The cartesian equation of C is an equation without the parameter t. Common techniques:

① **Make t the subject.** e.g. $x=2t-1, y=3t^2+e^t, t\in\mathbb{R}$

$$t=\frac{x+1}{2}$$

$$y=3\left(\frac{x+1}{2}\right)^2+e^{\frac{x+1}{2}}$$

② **Trigo identity formulas.** e.g. $x=2\cos t+1, y=3\sin t-2, 0\leqslant t\leqslant\pi$

$$\cos t=\frac{x-1}{2}, \quad \sin t=\frac{y+2}{3}$$

$$\cos^2 t+\sin^2 t=1$$

$$\frac{(x-1)^2}{4}+\frac{(y+2)^2}{9}=1, \quad y\geqslant -2$$

③ **Others.** e.g. $x=t-\frac{1}{t}, y=t+\frac{1}{t}, t\neq 0$

$$y+x=2t, \quad y-x=\frac{2}{t}$$

$$(y+x)(y-x)=4$$

$$y^2-x^2=4$$

Note: Avoid finding the cartesian eqn if the question did not ask you to.

3.5.2 Asymptotes

- Find (if possible) $t \to t_0$ such that $x \to x_0, y \to \pm\infty$ (vertical asymptote $x = x_0$), or $y \to y_0, x \to \pm\infty$ (horizontal asymptote $y = y_0$).
- Oblique asymptotes (if they exist) can only be found via the cartesian equation.

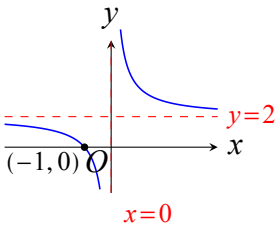
Example. $x = \dfrac{1}{t-1},\ y = 2t,\ t \neq 1$

As $t \to \pm\infty,\ x \to 0,\ y \to \pm\infty$

$\therefore x = 0$ is a vertical asymptote

As $t \to 1,\ x \to \pm\infty,\ y \to 2$

$\therefore y = 2$ is a horizontal asymptote



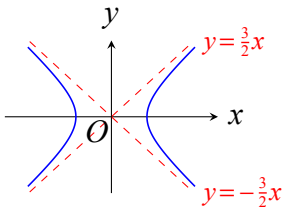
Oblique asymptotes. Cannot be found from the parametric form alone; find the cartesian equation first.

e.g. $x = 2\sec\theta,\ y = 3\tan\theta$

$$\sec^2\theta - \tan^2\theta = 1$$

$$\frac{x^2}{4} - \frac{y^2}{9} = 1$$

Oblique asymptotes: $y = \pm\dfrac{3}{2}x$



3.5.3 Intersections

Intersection between a parametric curve and another curve: sometimes both equations are needed.

e.g. $x = 2t + \ln t,\ y = t^2 - 1,\ t > 0$ and the circle $x^2 + y^2 = 25$:

$$(2t + \ln t)^2 + (t^2 - 1)^2 = 25$$

Using GC, $t = 0.00735$ or $t = 1.86$ (both > 0)

$\therefore$ intersections: $(-4.90,\ -1.00)$ and $(4.35,\ 2.47)$

This equation in t cannot be solved algebraically, so the GC is needed.

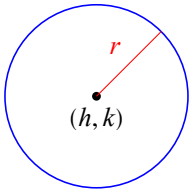
3.6 Conics

Circle

$$(x-h)^2+(y-k)^2=r^2$$

Centre: (h,k)

Radius: r

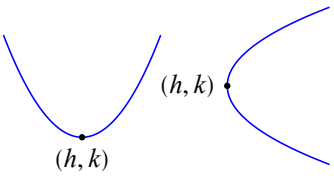


Parabola

$$y-k=a(x-h)^2 \qquad x-h=b(y-k)^2$$

$y-k=a(x-h)^2$:
turning point (h,k), $\dfrac{dy}{dx}=0$

$x-h=b(y-k)^2$:
turning point (h,k), $\dfrac{dy}{dx}$ undefined



Ellipse

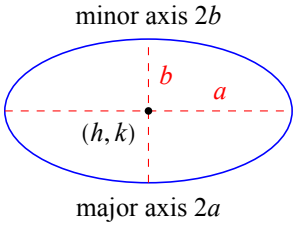
$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$$

Centre: (h,k)

"Horizontal radius" $=a$, "vertical radius" $=b$

For $a>b$: major axis horizontal (length $2a$, semi-major a); minor axis vertical (length $2b$, semi-minor b).

⋆ RHS ⇒ numerator coefficients of x^2, y^2 must all be 1.



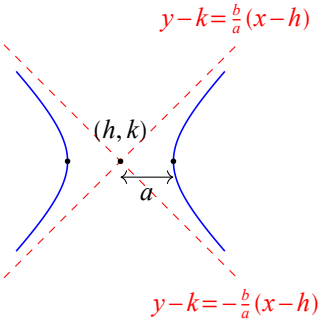
Hyperbola: Type I

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

Centre: (h, k)

Distance from centre to vertices $= a$

Asymptotes (dashed): $y - k = \pm\frac{b}{a}(x-h)$



- RHS $\Rightarrow$ numerator coefficients of x^2, y^2 must all be 1.
- Equation of asymptotes: $y - k = \pm\frac{b}{a}(x-h)$

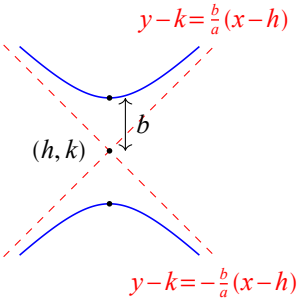
Hyperbola: Type II

$$\frac{(y-k)^2}{b^2} - \frac{(x-h)^2}{a^2} = 1$$

Centre: (h, k)

Distance from centre to vertices $= b$

Asymptotes (dashed): $y - k = \pm\frac{b}{a}(x-h)$



- RHS $\Rightarrow$ numerator coefficients of x^2, y^2 must all be 1.
- Equation of asymptotes: $y - k = \pm\frac{b}{a}(x-h)$
